

How to Use the SmartSchedule Programs

(v20251209)

Step 0: Preparations

1. Make sure you are placing the “SmartScheduling” folder in your AU OneDrive/SharePoint folder of your choice.
2. These are the subfolders and files you find in your “SmartScheduling” folder:

Subfolders	Description
0000_BuildingFiles	Here we find essential files that are used to build the “Don” sheet. A central file is the “source_DropDownMenus.xlsx” that is used to build DropDown menus for the “Don” sheet.
Step1_input_downloaded_AU_BannerPrevious	Here we dump the Banner report of a previous year from which we want to build the “Don” sheet.
Step2_input_downloaded_Seminary_Smart_Schedule	Here we dump the Internal Schedule (“Karen” sheet) from which we want to import courses into the new “Don” sheet.
Step3_input_downloaded_AU_BannerMostRecent	Here we dump the most recent Banner report so that it can be compared with the “Don” sheet.
Step3_output_DiscrepancyReport	Here the Banner report vs. “Don” sheet comparison outcomes are placed. Three files can be found here. The most important is the color coded “ColorReport SEM course updates.xlsx”.
x_SEM_CourseUpdate_LiveFolder_office365	Here the live “Don” sheet is living. Admins are interacting with the file (e.g. “DonSheet_202631” hosted by this folder.

Files	Description
• manual_SmartSchedule_programs.docx • manual_SmartSchedule_programs.pdf	These are docx and pdf version of the manual describing the programs.
• Run_Step1_SmartSchedule_NewDonSheet_creation.bat • Run_Step1_SmartSchedule_NewDonSheet_creation.command	Step1 programs. The step1 program generates the new “Don” sheet from a previous year Banner report. ⇒ The program with the ending “.bat” can be used in a Windows environment. ⇒ The program with the ending “.command” can be used in a MacOS environment. The new “Don” sheet is placed into the subfolder “x SEM CourseUpdate LiveFolder office365”
• Run_Step2_SmartSchedule_InternalSchedule-to-Don_import.bat • Run_Step2_SmartSchedule_InternalSchedule-to-Don_import.command	Step2 programs. The step2 programs import all courses from the Internal Schedule (“Karen” sheet) into the new “Don” sheet. ⇒ The program with the ending “.bat” can be used in a Windows environment. ⇒ The program with the ending “.command” can be used in a MacOS environment.

	The resulting file is found in the subfolder "x SEM CourseUpdate LiveFolder office365"
• Run_Step3_SmartSchedule_Don-Banner_comparison.bat • Run_Step3_SmartSchedule_Don-Banner_comparison.command	Step3 programs. The step3 program compare the most recent Banner report with the most recent "Don" sheet. ⇒ The program with the ending ".bat" can be used in a Windows environment. ⇒ The program with the ending ".command" can be used in a MacOS environment. The resulting files are found in the subfolder "Step3_output DiscrepancyReport"
• source_Step1_SmartSchedule_v20251208_NewDonSheet_creation.py • source_Step2_SmartSchedule_v20251208_InternalSchedule-to-Don_import.py • source_Step3_SmartSchedule_v20251208_Don-Banner_comparison.py	These three python programs (py) should not be deleted as they consist of the actual programs on which the bat and command files rely on.

3. Make sure that you do NOT touch/change/delete the contents of folder "0000_BuildingFiles". The folder contains the necessary files to build specific functionalities for the desired output files (Step1, Step2, Step3).
4. Make sure that the following folders are empty:
 - Step1_input_downloaded_AU_BannerPrevious
 - Step2_input_downloaded_Seminary_Smart_Schedule
 - Step3_input_downloaded_AU_BannerMostRecent
 - Step3_output_DiscrepancyReport
 - x_SEM_CourseUpdate_LiveFolder_office365

Step 1: Producing the NEW "SEM_course_updates" (the "Don Sheet") Spreadsheet from Banner

1. BEFORE you start the process: Close all Excel files that come from the SmartScheduling folder or one of its subfolders.
2. Download from the Banner website the Banner report for last year's semester you want to update.
3. Place the downloaded Banner report ("SWRSAUD") into the folder "Step1_input_downloaded_AU_BannerPrevious".
4. Double-click on the program called "Run_Step1_SmartSchedule" (a "bat" file for Windows users, a "command" file for MacOS users).
5. The folder "x_SEM_CourseUpdates_LiveFolder_office365" contains now the correctly formatted file with the name "New_SEM_course_updates_YYYYsemester.xlsx".
6. Rename the output file by adding the appropriate Year and semester information. For example:
 - "SEM_course_updates_2025-21_Spring.xlsx"
 - "SEM_course_updates_2025-31_Summer.xlsx"
 - "SEM_course_updates_2025-41_Fall.xlsx"

Step 2: Updating the NEW “SEM_course_updates” (the “Don Sheet”) from the Internal Schedule

1. BEFORE you start the process: Close all Excel files that come from the SmartScheduling folder or one of its subfolders.
2. (In case the Internal Schedule is used in the Google Office environment: Download the “Internal Schedule” [google sheet] and place it into the folder Step2_input_downloaded_Seminary_Smart_Schedule”).
3. Organize the tabs/sheets following the following sequence: Semester, Sync, Async (example: Sum 26 (main campus), SYNC 26-27, ASYNC 26-27). Save the changed file.
4. Double-click on the program called “Run_Step2_SmartSchedule” (a “bat” file for Windows users, a “command” file for MacOS users). You can find the updated “Don” sheet in the folder “x_SEM_CourseUpdate_LiveFolder_office365”.
5. Share the OneDrive/SharePoint link to this updated “Don” sheet with your collaborators.
6. The file can now be accessed through both the web browser and the Excel program installed on your computer. WARNING: make sure that when you use the Excel program that you are connected to the internet and have “AutoSave” on so that your changes are in sync with the web version.
7. Cancellations of courses need to be deleted manually in the Don sheet (Why?: Because there are many courses in the system that are not in the internal schedule but need to exist [e.g. dissertation proposal classes, independent studies, etc.]).

Step 3: Running Discrepancy reports between Banner and “SEM_course_updates” Spreadsheet

1. Download the most recent Banner version of the semester you are working on.
2. Place the downloaded Banner report file in folder “Step3_input_downloaded_AU_BannerMostRecent”.
3. Double-click on the program called “Run_Step3_SmartSchedule” (a “bat” file for Windows users, a “command” file for MacOS users).
4. When the program is finished you can see the Discrepancy report in folder “Step3_output_DiscrepancyReport”. The file “ColorReport_SEM_course_updates.xlsx” contains the color-coded differences.
5. You can, if you want copy/paste the data from “ColorReport_SEM_course_updates.xlsx” into your “SEM_course_updates_YYYY-S_semester.xlsx” file to help all admins to identify what needs to be worked on. HOWEVER, it is important that one copies the data by retaining the formatting (do a right-click on the selected rows you want to replace and choose “Paste Special/Formatting”).

You can repeat Step 3 as often as you need until there is a perfect match between Banner and the “Don” sheet.

A full video description can be found here: <https://youtu.be/ApfDT2q0lcw>